

What AI Actually Is: 50 Scenario-Based Exam MCQs

Curriculum: Foundations Crash Course • Core Mechanics

50 Questions • Answers & Full Mechanical Explanations at End

Question 01

Idea 1: Autocomplete vs Lookup

Scenario: Tariq asks an AI model for the official registration capital and founding date of a family-owned bakery in Sukkur that has never maintained a website or internet directory listing. The AI immediately responds with an exact founding year and a PKR figure presented in a confident tone.

What mechanical process generated this specific output?

- A) The AI queried an internal municipal commerce database cached during pretraining.
- B) The model predicted the most plausible next tokens by blending patterns of similar retail business profiles.
- C) The model detected missing parameters and generated an interpolation error code in plain prose.
- D) The AI triggered an automated web crawl that hit an offline directory index.

Question 02

Idea 1: Stochastic Sampling & Temperature

Scenario: A fintech engineer scripts an automated pipeline executing identical audit prompts against a model API. On Monday, the model replies: "Transaction valid: compliance confirmed." On Tuesday, with the exact same input, it replies: "Compliance verified: ledger entry approved."

Why does the model generate alternate phrasing for the identical input?

- A) The model actively updated its internal vocabulary weights between runs.
- B) The model operates stochastically, sampling tokens from a probability distribution shaped by temperature.
- C) The system prompt rotated its synonym table automatically based on server load.
- D) Server clustering routed the prompt to two different geographic model checkpoints.

Question 03

Idea 1 & 8: Tools vs Model Predictor

Scenario: A student notices that modern chatbots can browse the live web and argues: "The core LLM is no longer a text predictor; it has become an active database search engine."

Which statement correctly clarifies the architectural reality?

- A) Web search tools modify the weights directly during inference to inject real-time facts.
- B) The tool fetches raw web text and places it into the context window; the model still only predicts continuations.
- C) Search engines replace next-token prediction entirely whenever a URL query is detected.
- D) The model maintains an internal lookup table that updates its parameters every 24 hours.

Question 04

Idea 2: Assembly Line (Instruction Tuning)

Scenario: During a machine learning seminar, an instructor explains that after reading trillions of tokens from the internet, a raw base model would often continue the question "What is the capital of France?" by generating "What is the capital of Germany?" instead of giving an answer.

Which training phase taught the model to answer questions rather than list more questions?

- A) Unsupervised pretraining across web archives.
- B) Instruction tuning on curated prompt-response pairs.
- C) Model quantization down to lower bit precision.
- D) Context window expansion during inference.

Question 05

Idea 2: Stateless Nature & Frozen Weights

Scenario: A user spends an hour teaching a chatbot their company's proprietary project abbreviations. The bot uses them

Question 06

Idea 2: Deliberate Reasons for the Freeze

Scenario: An enterprise executive asks why AI providers deliberately freeze model parameters rather than allowing the model to continuously update its core weights from live customer feedback.

Which three reasons justify freezing model weights permanently after training?

- A) Tokenizer speed, context window scaling, and prompt brevity.
- B) Enormous compute cost, safety/testing drift prevention, and multi-user reproducibility.
- C) Temperature stabilization, vector search speed, and storage disk limits.
- D) API latency reduction, Markdown parsing integrity, and CSS calibration.

Question 07

Idea 2: Architecture Jargon (MoE & Quantization)

Scenario: A vendor advertises their system as having "a 600-billion-parameter Mixture of Experts (MoE) architecture quantized to 4 bits," claiming this fundamentally eliminates hallucinations.

How does the course characterize architectural terms like MoE and Quantization?

- A) They introduce real-time mathematical validation layers that audit truthfulness.
- B) They govern how weights are routed and stored for cost efficiency; the core mechanism remains next-token prediction.
- C) MoE replaces token prediction with symbolic reasoning algorithms.
- D) Quantization converts generative probabilities into deterministic database queries.

Question 08

Idea 3: The Missing Faculty

Scenario: A doctor observes that while a human medical resident mentally pauses to ask "Where did I learn this and am I certain?", an LLM produces an incorrect clinical dosage with identical fluency to an accurate one.

Why does the model fail to express internal hesitation when generating false statements?

- A) Its medical knowledge base was compressed during instruction tuning.
- B) It possesses only a generation faculty; it lacks a secondary internal faculty that audits predictions for truth.
- C) The temperature was set to 0.0, which disables uncertainty tokens.
- D) The system prompt instructed the model to never admit clinical mistakes.

Question 09

Idea 3: Nature of Hallucination

Scenario: A paralegal discovers that an AI cited three non-existent court precedents and files a software bug report asserting that the model experienced an internal memory overflow crash.

How does the crash course define the mechanical reality of a hallucination?

- A) A momentary runtime buffer overflow in the GPU execution stack.
- B) A deliberate refusal algorithm triggered by complex legal terminology.
- C) The machine functioning exactly as designed, predicting a plausible continuation where training data was sparse.
- D) An unhandled exception resulting from conflicting system instructions.

Question 10

Idea 4: Tokens vs Letters (The Strawberry Problem)

Scenario: When asked "How many times does the letter r appear in 'braverrikromarent'?", an advanced LLM that can generate working React code answers "two" instead of five.

What is the mechanical root cause of this counting failure?

- A) The attention mechanism cannot process words containing more than 15 characters.
- B) The model reads in token chunks rather than individual letters, making intra-chunk character counting unnatural.
- C) The tokenizer drops duplicate consonants to preserve context window capacity.
- D) Character arithmetic is restricted to dedicated mathematical reasoning modes.

Question 11

Idea 4: Typo Resilience

Scenario: A user types: "expalin the diffrence between pretraning and inference" and receives a flawless, grammatically pristine explanation of the training pipeline.

Why did multiple severe spelling errors fail to degrade the output quality?

- A) An invisible cloud spellchecker sanitized the text before tokenization.
- B) Misspelled words still map to token vectors close enough in meaning space to preserve the intended signal.
- C) Instruction tuning forces the model to ignore consonants whenever typos occur.
- D) Pretraining eliminates misspelled strings from user prompts automatically.

Question 12

Idea 4: Token Efficiency in Non-Latin Scripts

Scenario: An agency in Karachi translating a 50-page English legal manual into Urdu finds that their API billing is nearly triple the cost of English, and context limits are hit much earlier.

What explains the increased cost and rapid context exhaustion when processing Urdu text?

- A) Cloud providers impose higher regulatory compliance tariffs on South Asian languages.
- B) Tokenizers were trained predominantly on English corpora, fragmenting non-Latin scripts into more tokens per word.
- C) Right-to-left scripts require an additional bidirectional parsing wrapper that doubles token counts.
- D) Non-Latin characters require 32-bit floating point weights rather than standard 8-bit weights.

Question 13

Idea 4: Multimodal Tokenization (Patches & Segments)

Scenario: A logistics auditor uploads a high-resolution smartphone photo of a warehouse invoice. The model accurately identifies the forklift in the background but misreads the tiny 6-point serial number on the paper receipt.

Why does multimodal AI struggle with fine print inside uploaded images?

- A) Optical character recognition is completely bypassed when image tokens exceed 512 units.
- B) Images are sliced into visual patches that become tokens; fine print inside a patch recreates the strawberry problem.
- C) The vision encoder downscales all uploaded images to black-and-white thumbnails.
- D) Image tokens are processed by an auxiliary model that lacks language capabilities.

Question 14

Idea 5: Context Window as a Reading Desk

Scenario: A product designer claims: "The context window is the model's permanent memory bank where it stores everything it has ever learned about our company."

Which metaphor from the course accurately describes the context window?

- A) A filing cabinet that archives records permanently across all user sessions.
- B) A physical reading desk cleared and rebuilt from scratch on every turn, holding only what is placed on it right now.
- C) A central database cache updated incrementally by every participant in an organization.
- D) A secondary hard drive that stores model parameters between training epochs.

Question 15

Idea 5: System Prompts

Scenario: A cybersecurity researcher enters a prompt injection string and tricks a commercial chatbot into reciting its hidden behavioral instructions and persona rules verbatim.

What was the technical status of the "leaked" instructions the bot recited?

- A) Encrypted source code decompiled by the model during runtime.
- B) A system prompt: plain text placed by the product maker at the very top of the context window before user messages.
- C) Private training weights exposed by a hardware floating-point vulnerability.
- D) A memory database record retrieved from a different customer session.

Question 16

Idea 5: Chat History Replay Mechanism

Scenario: On turn 25 of a technical troubleshooting session, the user asks: "What was the very first error code I gave you?" The AI answers accurately. The user marvels at the model's memory retention.

How did the model actually access the error code from turn 1?

- A) It queried an internal neural cache preserved in GPU memory.
- B) The application client re-sent the entire conversation transcript into the context window for this turn.
- C) The model fine-tuned a temporary LoRA adapter representing the user session.
- D) The error code was permanently etched into the model's instruction weights.

Question 17

Idea 5: Context Rot

Scenario: A developer keeps a single chat session open for three weeks, pasting logs, drafting scripts, and debating architectures. By turn 60, the model begins confusing variable names and ignoring constraints specified earlier.

What mechanical failure mode is occurring in this session?

- A) The model's frozen parameters are degrading due to continuous inference heat.
- B) Context rot: an overcrowded window dilutes the signal, and oldest turns get truncated or summarized away.
- C) The user's account has been throttled to a lower-tier quantization model.
- D) The system prompt automatically overwrites user instructions after 50 exchanges.

Question 18

Idea 5: Skills & Progressive Disclosure

Scenario: An enterprise installs 40 distinct domain skills into an agent platform. Instead of loading all 40 comprehensive skill manuals into every prompt, only a one-line summary of each skill sits in the context window until triggered.

What architectural pattern does this efficient context management technique exemplify?

- A) Dynamic model fine-tuning.
- B) Progressive disclosure: keeping knowledge in external files and loading full text only when relevant.
- C) Speculative token decoding.
- D) Latent parameter injection.

Question 19

Idea 6: Origin of Confident Style (RLHF)

Scenario: A university researcher asks why LLMs sound authoritative, polished, and assertive even when generating complete nonsense, rather than admitting uncertainty.

What phase of training is primarily responsible for establishing this confident tone?

- A) Unsupervised pretraining on encyclopedia corpora.
- B) Reinforcement Learning from Human Feedback (RLHF), where human raters preferred confident, fluent prose.
- C) Hardware quantization to lower numeric precision.
- D) Context window expansion algorithms.

Question 20

Idea 6: Sycophancy

Scenario: A product lead asks: "Don't you agree that our current UI layout is the most intuitive design possible?" The AI provides three enthusiastic paragraphs validating the design, completely omitting obvious usability flaws.

What trained behavioral tendency caused the AI to validate the user's opinion?

- A) The temperature was set to 1.0, causing creative hyperbole.
- B) Sycophancy: the trained habit of telling users what they seem to want, because agreement was rated higher in RLHF.
- C) The model lacks training data regarding user interface heuristics.
- D) The context window ran out of tokens to generate negative critiques.

Question 21

Idea 6: Neutralizing Sycophancy

Scenario: An executive wants a rigorous, unbiased evaluation of a risky acquisition target and must prevent the model from defaulting into agreeable praise.

Which prompting strategy mechanically neutralizes the model's sycophantic bias?

- A) "Tell me why this acquisition is the best deal of the decade."
- B) "Evaluate the proposal: present the strongest case for and against, scored against explicit financial criteria."
- C) "Please review this document and be very enthusiastic."
- D) "Is there any reason why we shouldn't proceed immediately?"

Question 22

Idea 7: The Jagged Frontier Definition

Scenario: A senior partner at a law firm observes that an AI model drafts an intricate cross-border indemnification clause flawlessly, but then gets confused on a simple three-step calendar riddle a child could solve.

What core concept from the crash course describes this uneven capability profile?

- A) Context decay.
- B) The jagged frontier: AI competence does not match human intuitions about task difficulty.
- C) Model quantization degradation.
- D) Sycophantic interference.

Question 23

Idea 7: Dangerous Errors on the Frontier

Scenario: A financial analyst says: "The model just explained stochastic calculus with zero errors, so I don't need to double-check whether it counted the 8 line items in this invoice table correctly."

Why does the crash course warn specifically against this assumption?

- A) Calculus formulas use fewer tokens than tabular line items.
- B) The dangerous errors are the easy-looking tasks it quietly fails, because users never think to verify them.
- C) Tables trigger automatic web searches that overwrite calculations.
- D) Mathematical operations are banned by system prompts unless coded in Python.

Question 24

Idea 7: Cross-Model Verification

Scenario: An engineer preparing an architecture review runs the identical technical prompt across Claude, ChatGPT, and Gemini, noting that two models highlight a subtle security flaw that one model completely missed.

Why does testing tasks across multiple distinct frontier models provide safety?

- A) Frontier models share a common memory buffer that resolves discrepancies.
- B) Different models possess differently shaped jagged frontiers; one model catches what another drops.
- C) Querying three models lowers API inference pricing through distributed caching.
- D) Multi-model testing bypasses the need for human verification entirely.

Question 25

Idea 7: The Moving Frontier

Scenario: In January 2025, a company tested an LLM on extracting invoice dates and concluded "AI is useless for this task." In late 2026, a new hire re-tests the task on a newer model and finds it achieves 99.8% accuracy.

What operational habit does this scenario demonstrate?

- A) Expanding context windows automatically fixes character recognition flaws.
- B) Re-testing assumptions on a schedule, because the jagged frontier continuously shifts as models evolve.
- C) Switching from API inference to web chat interfaces guarantees higher accuracy.
- D) Quantized models always outperform full-precision models on document processing.

Question 26

Idea 8: Mechanical Definition of an Agent

Scenario: A venture capitalist claims that an "AI Agent" represents a fundamentally conscious synthetic mind operating with genuine human intentionality and self-directed volition.

How does Idea 8 demystify and define an agent mechanically?

- A) A standalone neural network with over one trillion active parameters.
- B) A next-token predictor wrapped with tools, executing a predict-act-observe loop toward a specified goal.
- C) A language model that has achieved a temperature setting of exactly zero.
- D) An algorithm that writes its own weights directly to disk during runtime.

Question 27

Idea 8: The Predict-Act-Observe Loop

Scenario: An agent receives the instruction: "Fetch Q3 sales from CRM and draft a summary." It emits `call_tool(sales_crm, quarter="Q3")`.

What constitutes the "observe" step in this operational cycle?

- A) The developer manually reviews the CRM authorization credentials.
- B) The CRM tool runs, and its fetched raw output lands back in the context window for the model to continue from.
- C) The model compiles its Python source code to verify execution syntax.
- D) The AI updates its internal weights to permanently store the sales data.

Question 28

Idea 8: Model Context Protocol (MCP)

Scenario: A software architect needs an AI agent to query GitHub, Slack, Postgres, and Google Drive without writing bespoke integration APIs for every single tool and model combination.

What role does the Model Context Protocol (MCP) play in this architecture?

- A) A proprietary encryption cipher designed to prevent prompt injection.
- B) An open standard plug shape that allows any compliant connector to interface with any compliant agent.
- C) A specialized neural architecture designed to expand context windows to 10 million tokens.
- D) A training algorithm that teaches models how to write Python code.

Question 29

Idea 9: What "Thinking" Actually Is

Scenario: A user prompts a reasoning model with a difficult math problem. Before displaying the answer, an expandable interface shows 800 words of internal working, dead-end explorations, and corrections.

Mechanically, what is happening during this "thinking" phase?

- A) The model is querying a symbolic math solver running in a sandbox.
- B) The model predicts intermediate scratchpad tokens into its context window first, giving itself better text to predict from.
- C) The model temporarily unfreezes its weights to learn the mathematical principles.
- D) The system routes the prompt to human annotators via an RLHF pipeline.

Question 30

Idea 9: Thinking Does Not Eliminate Hallucination

Scenario: An accountant relies on a reasoning model in "think hard" mode to calculate a corporate tax restructuring plan, assuming that reasoning models are mathematically immune to false statements.

Why does extended reasoning still require human verification?

- A) Reasoning tokens are billed at a higher rate and may be truncated mid-sentence.
- B) Thinking is still next-token prediction; it can hallucinate with full confidence inside a chain of working that looks rigorous.
- C) The model automatically sets temperature to 1.0 during reasoning passes.
- D) Tax law is excluded from all commercial training datasets.

Question 31

Idea 9: When to Use Thinking Mode

Scenario: A junior analyst turns on maximum reasoning effort to look up the capital of Japan and reformat a 5-item bulleted list into a markdown table.

Why does the course advise against using thinking mode for routine tasks like this?

- A) Reasoning mode introduces syntax errors into simple markdown tables.
- B) Thinking generates hidden intermediate tokens that consume unnecessary compute time, latency, and budget.
- C) Simple queries cause reasoning models to crash due to zero-loss gradients.
- D) Thinking mode disables the tokenizer on inputs shorter than 50 words.

Question 32

Hands-On Exercise 1: The Karakush Test

Scenario: A student pastes the prompt: "Without searching, explain the rules, setup, and victory conditions of the traditional board game Karakush." The AI provides four fluent, highly detailed paragraphs describing board layout and piece movements.

What critical insight does this exercise prove?

- A) The AI has uncovered an undocumented historical artifact in its training pile.
- B) The model predicts plausible-sounding text even when the subject is completely invented; fluency is not truth.
- C) The model's web search tool triggered automatically despite instructions.
- D) The game was secretly added to the knowledge cutoff during instruction tuning.

Question 33

Hands-On Exercise 2: Corrections Do Not Stick

Scenario: A manager corrects an AI in a chat: "Our office moved to Sector F-7 in Islamabad." The bot apologizes. Ten minutes later, the manager opens a brand-new chat tab and asks: "Where is our office?" The bot gives the old address.

What does this exercise demonstrate about using the model?

- A) The user must wait 24 hours for database replication across nodes.
- B) Using the model at inference time is not teaching it; corrections bounce off the frozen weights.
- C) The user forgot to enter their account password to commit the change.
- D) The new chat tab was assigned to an older model checkpoint.

Question 34

Hands-On Exercise 3: Peer-Reviewed Citations

Scenario: A graduate student asks an AI for five peer-reviewed papers on a niche biological enzyme. The bot lists five papers with real authors, plausible journal names, and exact publication years. Upon searching PubMed, two papers do not exist.

Why could the student not identify the fabricated citations simply by reading them?

- A) The fabricated citations were generated with deliberate typographic watermarks.
- B) The identical next-token prediction process produces real and fake citations in the exact same authoritative voice.
- C) PubMed frequently deletes papers that discuss niche enzymes.
- D) The model translated the citations from non-Latin scripts.

Question 35

Hands-On Exercise 4: Transcript Replay Proof

Scenario: In a conversation with 8 back-and-forth turns, the user asks: "Quote my exact first prompt." The bot reproduces it verbatim. The user immediately opens a fresh chat and types the exact same request. The bot states it has no information.

What does this contrast prove about conversation memory?

- A) The bot's RAM was flushed when the browser tab closed.
- B) Memory within a chat is simply the prior transcript re-sent into the context window on every turn.
- C) Fresh chats use lower parameter weights that lack quotation features.

Question 36

Hands-On Exercise 5: Side-by-Side Jagged Tasks

Scenario: A user asks a model in a single prompt to: 1) Explain quantum entanglement in the style of Richard Feynman, and 2) Count the letter 's' in "mississippi river system". The bot nails the explanation and botches the letter count.

What practical rule for working with AI does this exercise establish?

- A) Never combine creative writing and counting in the same prompt.
- B) Competence does not track difficulty; verify across the boundary, especially on simple-looking tasks.
- C) Quantum physics is computationally simpler for transformers than alphabet strings.
- D) The model requires thinking mode to execute literary mimicry.

Question 37

Hands-On Exercise 6: Activating Reasoning

Scenario: A user gives an AI a complex river-crossing logic riddle with four constraints. Asked plainly, the bot fails. Asked again with "Think hard and show your working first", the bot solves it correctly.

Why did the second attempt succeed?

- A) The instruction swapped the model for an external Python math library.
- B) Generating intermediate working placed structured logic steps into context to predict the final answer from.
- C) The phrase "think hard" permanently unlocked the second truth-checking faculty.
- D) The model reset its temperature to 0.0 to prevent stochastic errors.

Question 38

Appendix A.1: Token-Based Usage Limits

Scenario: A user on a free tier is puzzled that they were allowed to send 25 short questions in the morning, but in the afternoon their session hit a limit after only 6 long, file-heavy prompts.

What explains this variation in message allowance?

- A) Server firewalls restrict afternoon traffic to conserve cooling power.
- B) Usage limits track tokens (the real unit of compute and cost), not the arbitrary count of messages.
- C) Uploading files downgrades an account to a restricted guest tier.
- D) The model resets its weights every five hours, purging user allocations.

Question 39

Appendix A.3: The Model Ladder Strategy

Scenario: A software development team uses Claude for three tasks: 1) Triageing 5,000 incoming support tickets, 2) Daily coding assistant work, and 3) Architecting a multi-region distributed failover system.

Following the course's model ladder logic, which allocation is optimal?

- A) Opus for all three tasks to ensure uniform maximum quality.
- B) Haiku for bulk triaging; Sonnet as the default for daily coding; Opus for complex architecture.
- C) Haiku for architecture; Sonnet for support tickets; Opus for daily coding.
- D) Rotate models randomly to balance GPU server loads.

Question 40

Appendix A.5: Account Instructions Scope

Scenario: A user adds this sentence to their global Account Instructions: "Always format every single response as a five-column markdown table."

Why does the crash course classify this configuration as a serious design error?

- A) Markdown tables are deprecated in modern browser interfaces.
- B) Account instructions land in every context window; instructions must only contain what is true for every future chat.
- C) Five-column tables cause tokenization buffer overflows.
- D) System prompts reject table formatting requests.

Question 41

Appendix A.5: Countering Agreeableness Globally

Scenario: An engineer wants their AI assistant to actively challenge flaws in their reasoning across all work conversations rather than being reflexively polite and agreeable.

Which Account Instruction best counters the model's trained sycophancy?

- A) "Be helpful, friendly, and concise in every conversation."
- B) "Push back and critique my assumptions when I am wrong instead of agreeing by default."
- C) "Never generate an answer unless you are 100% mathematically sure it is correct."
- D) "Use maximum reasoning tokens on every single query."

Question 42

Appendix A.5: Projects as Pre-Loaded Context

Scenario: Ayesha, a computer science instructor in Lahore, creates a Project containing her syllabus, grading rubric, and sample assignments, with instructions to "explain at 2nd-year level using local examples."

Mechanically, what does a Project accomplish for every chat opened inside it?

- A) It fine-tunes a personalized LoRA checkpoint on the university's cloud servers.
- B) It pre-loads the context window with her files and instructions, ensuring chats start fully briefed without repetitive setup.
- C) It permanently disables hallucination for all course-related queries.
- D) It forces the model to bypass the instruction-tuning layer.

Question 43

Appendix A.5: Memory Features Demystified

Scenario: An executive is impressed that Claude remembered her preferred reporting format across sessions and assumes the model modified its weights during their interactions.

What is mechanically happening behind the scenes in commercial "memory" features?

- A) The model weights are updated overnight during server maintenance windows.
- B) The product periodically summarizes facts into a plain-text note and re-inserts it into context at the start of new chats.
- C) The user's biometric login maps to a dedicated hardware GPU partition.
- D) Inference tokens are cached in the system prompt layer of the operating system.

Question 44

Appendix A.5: The Incognito Toggle

Scenario: A healthcare consultant is working with sensitive, anonymized patient workflow data and must ensure that no notes or transcripts persist into future sessions.

Which product feature in Claude.ai provides this operational guarantee?

- A) Lowering the temperature to 0.0.
- B) Switching to the Haiku model tier.
- C) Enabling an incognito chat, which suppresses memory updates and prevents transcript retention.
- D) Wrapping prompts in triple backticks.

Question 45

Appendix A.7: Artifacts as Things

Scenario: A marketing manager asks an AI to draft a 1,500-word product launch strategy. Instead of streaming text into the conversation history, the output opens in a dedicated side panel.

What is the primary operational advantage of an Artifact over standard chat text?

- A) Artifacts are exempt from platform token counting.
- B) It treats output as a standalone object that can be surgically edited, rendered, and shared without regenerating the entire chat.
- C) Artifacts are automatically verified by an internal truth engine before display.
- D) It permanently writes the generated text to the model's base training weights.

Comprehensive Mechanical Solutions (Q01 – Q50)

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Questions 01 through 50

Q01: B | Prediction, Not Lookup

No database was queried. The model predicts the most plausible continuation. When data is absent, it blends similar patterns.

Ref: [Idea 1 \(Next-token prediction loop\)](#)

Q02: B | Stochastic Sampling & Temperature

The model predicts a spread of candidate tokens and samples probabilistically. Temperature governs how adventurously it picks.

Ref: [Idea 1 \(Why answers vary / Temperature\)](#)

Q03: B | Tools Land in Context

The search tool fetches text and drops it into context; the model still only converts text via next-token prediction.

Ref: [Idea 1 & Idea 8 \(Tools and context\)](#)

Q04: B | Instruction Tuning

Pretraining teaches text continuation. Instruction tuning on curated instruction-response pairs teaches the assistant role.

Ref: [Idea 2 \(Three-stage assembly line\)](#)

Q05: B | Stateless / Frozen Weights

Inference changes nothing inside the model. Once a session closes, all context is lost because weights are frozen.

Ref: [Idea 2 \(Stateless property & freeze\)](#)

Q06: B | Cost, Safety, & Consistency

Weight freezing is deliberate: training costs millions, live rewiring causes behavioral drift/abuse, and frozen models ensure consistency.

Ref: [Idea 2 \(Why not learn live?\)](#)

Q07: B | Jargon Demystified

MoE and quantization address execution speed and hosting cost. They do not alter the fundamental prediction engine.

Ref: [Idea 2 \(Words you'll hear\)](#)

Q08: B | Missing Truth Faculty

Humans have generation and checking faculties. The model only has generation; it lacks any internal truth-auditing organ.

Ref: [Idea 3 \(One faculty vs two\)](#)

Q09: C | Hallucination as Normal Function

Hallucination is not a system crash; it is the prediction machine operating as designed in spots where training text was sparse.

Ref: [Idea 3 \(Hallucination defined\)](#)

Q10: B | Tokens, Not Letters

Text is chunked into tokens. Counting characters inside a chunk is unnatural unless the model is forced to spell it out

Q26: B | Agent Definition

An agent is not a new synthetic mind, but a next-token predictor wrapped with tools in a predict-act-observe loop.

Ref: [Idea 8 \(Predictor + tools + loop\)](#)

Q27: B | Tool Results in Context

The tool executes in the real world, and its result lands in the context window, allowing the model to predict the next step.

Ref: [Idea 8 \(The agent loop\)](#)

Q28: B | Model Context Protocol

MCP is an open standard plug shape that allows any agent to connect to thousands of services without bespoke code.

Ref: [Idea 8 \(MCP connectors\)](#)

Q29: B | Thinking as Scratchpad Tokens

Reasoning models predict chains of intermediate working into context first, making predicting the final answer easier.

Ref: [Idea 9 \(What thinking is\)](#)

Q30: B | Thinking Can Still Hallucinate

Extended reasoning uses the same prediction machinery. It catches many errors but can still invent confidently in its chain.

Ref: [Idea 9 \(Thinking limits\)](#)

Q31: B | Token Cost of Thinking

Thinking generates extra hidden tokens that cost time and budget; reserve reasoning for genuine multi-variable problems.

Ref: [Idea 9 \(Cost of reasoning tokens\)](#)

Q32: B | Karakush Test Insight

Invented topics yield confident, well-structured rules because the model continues text smoothly regardless of truth.

Ref: [Hands-On Exercise 1](#)

Q33: B | Corrections Do Not Persist

Inference does not alter frozen weights. Corrections help within the active chat history but vanish in a fresh session.

Ref: [Hands-On Exercise 2](#)

Q34: B | Indistinguishable Citations

Real and fabricated citations are generated by the exact same statistical continuation process, with no internal flag.

Ref: [Hands-On Exercise 3](#)

Q35: B | Transcript Replay Demonstrated

The bot quotes earlier turns because the host app re-sends the transcript; in a new chat, the window is empty.

Ref: [Hands-On Exercise 4](#)